

DTA0611 - Machine Learning

CO1 - AT2

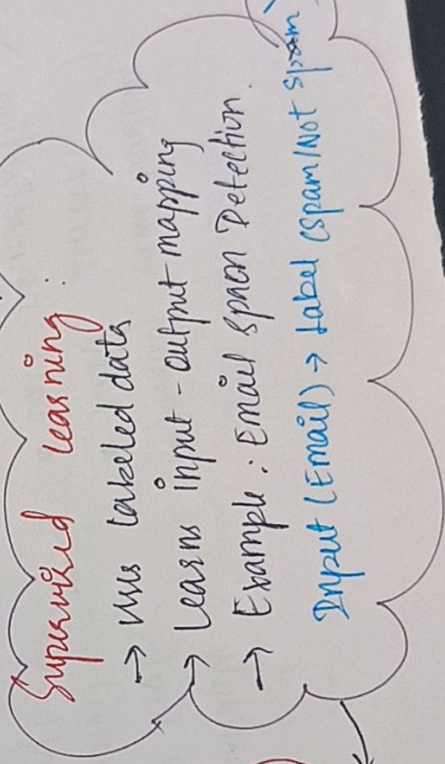
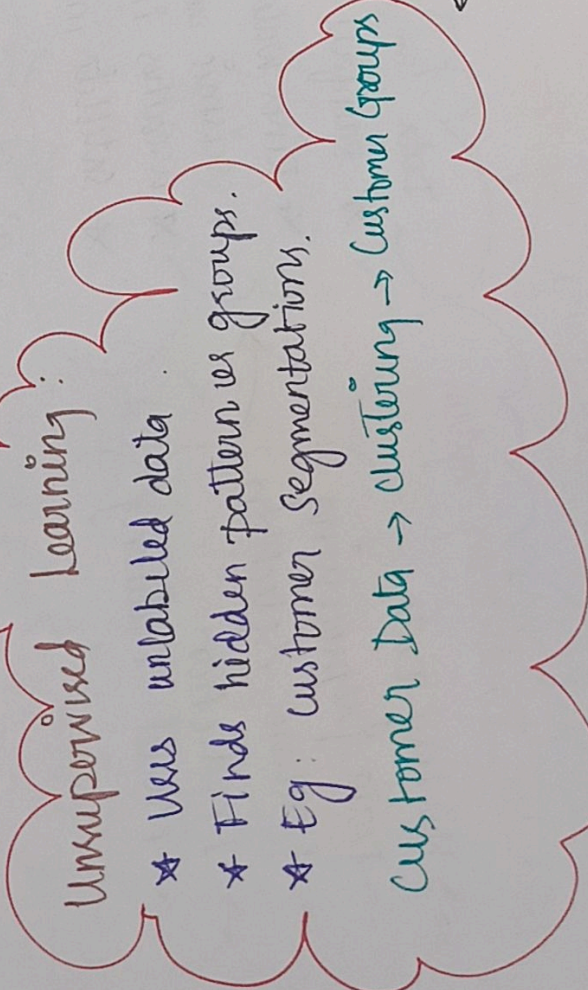
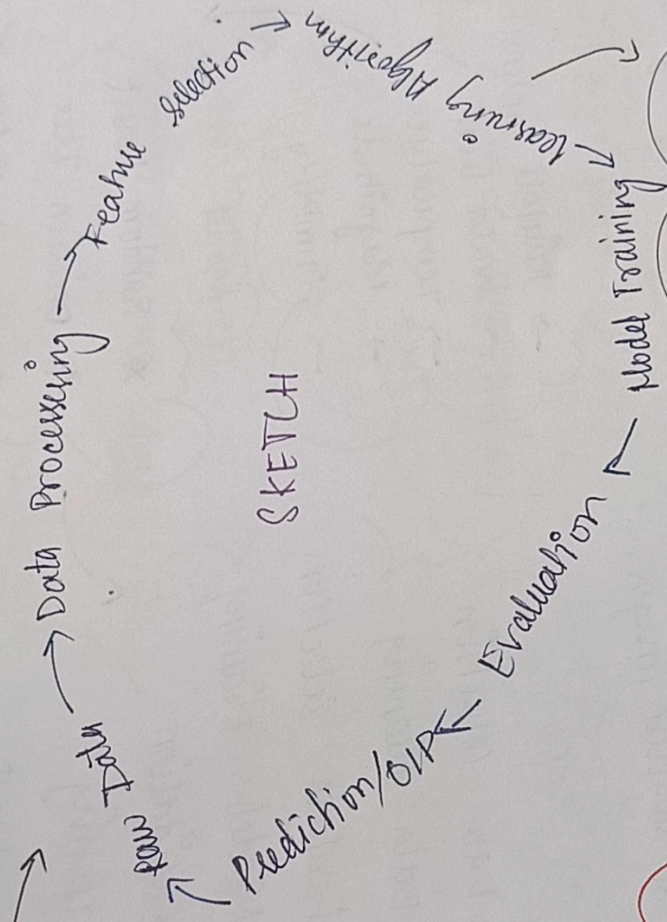
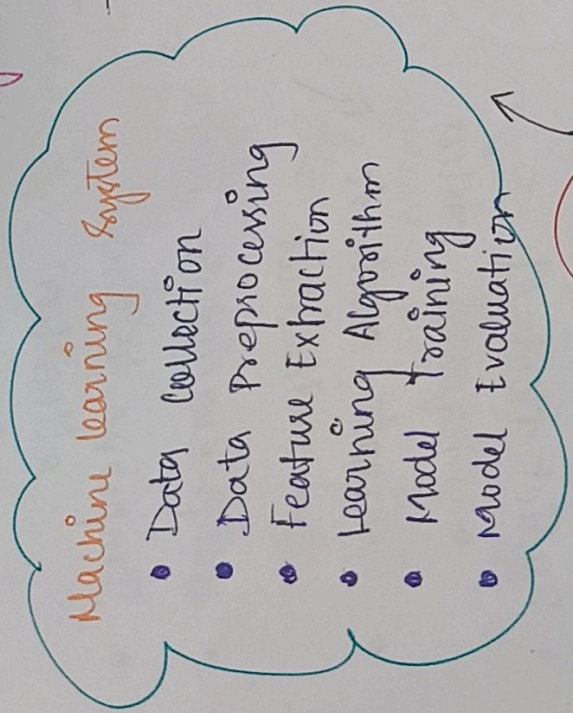
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1. Machine Learning System - Blocks and Learning Methods



2. Machine Learning System for Disaster Management

Disaster Prediction System

- History Data
- Data Preprocessing
- Feature Selection
- Prediction
- Model Training

Data sources

- * Rainfall records
- * River water levels
- * seismic activity
- * weather info
- * satellite images.

Data Preprocessing

- Missing value handling
- Noise removal
- Data normalization

Feature Selection

- Rainfall intensity
- Water level
- Temperature
- Magnitude
- Humidity

Model choice

- * Random Forest
- * Decision Tree
- * Neural Network

a) Steps:

- Data collection
- Data cleaning
- Feature selection
- Model Training
- validation
- Testing
- Evaluation.

b) Emergency Assistance

- Predict floods & earthquakes earlier
- Generate alerts to authorities
- Help evacuation plans
- Improve resource allocation

Disaster Prediction - Tennis Dataset

3. Find - S Algorithm - Tennis Dataset

Positive Example

Output	Temp	Humidity	Wind	Play
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Overcast	Mild	Normal	Strong	Yes

Final Hypothesis

$$H = \langle ?, ?, ?, ? \rangle$$

Step: 1

First Positive Example

$$S_1 = \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$$

Step: 2

Compare with Example 4

$$S_2 = \langle ?, ?, ?, \text{High}, \text{Weak} \rangle$$

Step: 3

Compare with examples

$$S_3 = \langle ?, ?, ?, \text{Weak} \rangle$$

Step: 4

Compare with Example 7

$$S_4 = \langle ?, ?, ?, ? \rangle$$

4. Candidate Elimination Algorithm - Enjoy Sport Dataset

Initialization

Specific Boundary:

$S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

General Boundary:

$G_0 = \langle ?, ?, ?, ? \rangle$

Final version space

$S = \langle \text{Sunny}, ?, ?, \text{Weak} \rangle$

$G_1 = \langle \text{Sunny}, ?, ?, ? \rangle$

Example 5 (Negative)

Rainy, Warm, Normal, Strong

No change because G_1 already

concludes Rainy.

Example 1 (Positive)

$S_1 = \langle \text{Sunny}, \text{Warm}, \text{Normal}, \text{Weak} \rangle$

$G_1 = \langle ?, ?, ?, ? \rangle$

Example 2 (Positive)

$S_2 = \langle \text{Sunny}, \text{Warm}, ?, ? \rangle$

$G_2 = \langle ?, ?, ?, ? \rangle$

Example 3 (Negative)

Rainy, Cold, High, Weak

Specialize G_1 :

$G_3 = \{$

Remove Inconsistent Hypotheses:

$G_4 = \{$

$\langle \text{Sunny}, ?, ?, ? \rangle$

$\rangle$

Product Dataset

Buy

Customer

Find - S Algorithm -

5. Find - S - Algorithm - Customer

Positive Examples

Age	Income	Student	Credit	Buy
Middle	High	No	Fair	Yes
Senior	Medium	No	Fair	Yes

Buy Product Dataset

Step: 1

First Positive Example

$S_1 = \langle \text{Middle, High, No, Fair} \rangle$

Step 2

Compare with Second Positive Example

Age: Middle $\neq$ Senior $\rightarrow ?$

Income: High $\neq$ Medium $\rightarrow ?$

Student: No = No $\rightarrow$ No

Credit: Fair = Fair $\rightarrow$ Fair

Final Hypothesis:

$H = \langle ?, ?, ?, \text{No, Fair} \rangle$

Interpretation:

A customer is likely to buy the product if:

- Student = No

- Credit Rating = Fair

- Age and income can be any value.